ResiDual for Audio: Spectral Reweighting of Residual Streams in CLAP Models

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Abstract

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1. Method

A. CLAP and HTS-AT: Architecture, Notation, and Residual Decomposition

This appendix is a self-contained reference for the two components at the core of our system. We begin by describing the CLAP dual-encoder model and its configuration (§A.1). We then present the HTS-AT audio backbone in detail, covering input preprocessing, hierarchical stage structure, and the internal mechanics of each attention block (§A.2). We then derive the per-head residual decomposition that underlies our analysis (§A.3). Finally, we provide implementation details for the ResiDual spectral reweighting (§C.1). All notation is introduced inline and collected in Table 3 for reference.

A.1. CLAP Overview

CLAP (Contrastive Language–Audio Pretraining) (**?**) is a dual-encoder model that aligns audio and text representations in a shared embedding space via contrastive learning. The audio encoder is HTS-AT (**?**) and the text encoder is GPT-2 (**?**) (embedding dimension 768), whose weights are frozen throughout training. Both encoders are independently projected to a joint space of dimension $d_{\text{proj}} = 1024$ via dedicated linear projection heads; similarity is measured by temperature-scaled cosine similarity with InfoNCE loss at temperature $\tau = 0.003$. Zero-shot audio classification is performed by comparing an audio embedding against the embeddings of textual class prompts.

The full configuration used in this work is reported in Table 1.

Table 1. CLAP configuration parameters.

Parameter	Value
Text encoder	GPT-2
Text encoder embedding dim	768
Audio encoder	HTS-AT
Audio encoder output dim (D_3)	768
Joint projection dim (d_{proj})	1024
Contrastive temperature τ	0.003
Sampling rate	44 100 Hz
Audio duration	7 s
Mel bands	64
FFT window	1024
Hop size	320
$f_{\min}$ / $f_{\max}$	50 Hz / 8 000 Hz

A.2. HTS-AT Architecture

HTS-AT (**?**) is a Swin-Transformer variant adapted for audio spectrograms. Its computation is organised into four hierarchical *stages*, each composed of several *blocks*, as illustrated in Figure 1. We describe the pipeline from raw audio to the final embedding in order: input preprocessing, patch embedding, stage structure, and the attention mechanism inside each block.

Input preprocessing and patch embedding. Each audio clip is converted to a 64-band log-mel spectrogram ($f_{\min} = 50$ Hz, $f_{\max} = 8{,}000$ Hz, STFT window 1024, hop 320) and normalised per mel-band. The resulting time–frequency matrix is rearranged into a square image $\mathbf{x}_{\text{img}} \in \mathbb{R}^{1\times 256\times 256}$ by folding the time axis into four contiguous segments of 256 frames stacked vertically over the 64 mel-frequency bands ($4\times 64 = 256$ rows, 256 columns). This image is then split into 4×4 non-overlapping patches via a strided `Conv2d` layer (kernel 4×4, stride 4), followed

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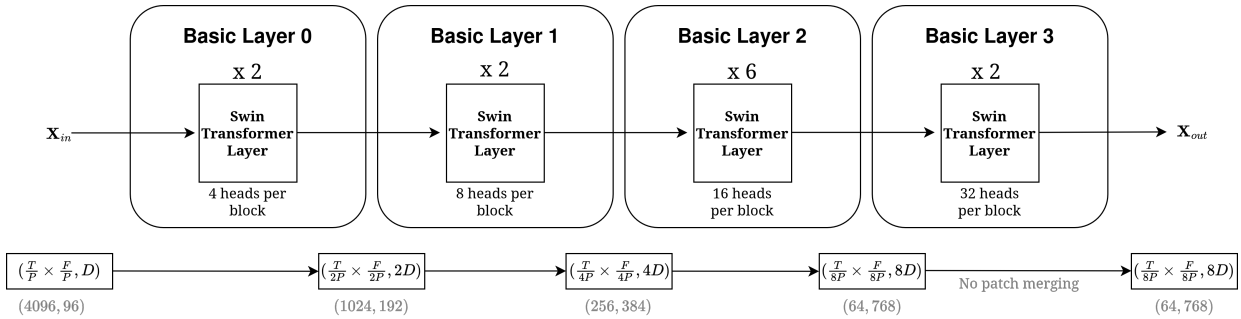


Figure 1. HTS-AT hierarchical architecture. The model consists of four stages of increasing embedding dimension: Stage 0 (2 blocks, 4 heads), Stage 1 (2 blocks, 8 heads), Stage 2 (6 blocks, 16 heads), and Stage 3 (2 blocks, 32 heads). PatchMerging is applied after Stages 0, 1, and 2, halving the spatial side length and doubling the feature dimension at each transition. Stage 3 has no PatchMerging. The input token sequence has spatial size $64 \times 64 = 4{,}096$ tokens with embedding dimension $D_0 = 96$; after three PatchMerging operations the final spatial size is $8 \times 8 = 64$ tokens with $D_3 = 768$.

by `LayerNorm`, yielding the input token sequence

$$\mathbf{Z}^{(0,0)} = \mathbf{X}_{\text{in}} \in \mathbb{R}^{4096\times 96}, \tag{1}$$

where $4096 = 64 \times 64$ tokens each have embedding dimension $D_0 = 96$.

Stage structure and PatchMerging. The four stages operate at progressively coarser spatial resolutions, as summarised in Table 2. We index the residual stream as $\mathbf{Z}^{(\ell,b)}$, where $\ell \in \{0, 1, 2, 3\}$ is the stage index and $b \in \{1, \ldots, B_\ell\}$ is the block index within that stage.

After Stages 0, 1, and 2, a *PatchMerging* layer concatenates each 2×2 neighbourhood of spatially adjacent tokens into a single vector of dimension $4D_\ell$, then projects it to $2D_\ell$ via a bias-free linear layer. This halves the spatial side length S_ℓ and doubles the embedding dimension:

$$\mathbb{R}^{S_\ell^2 \times D_\ell} \xrightarrow{\text{PatchMerging}} \mathbb{R}^{(S_\ell/2)^2 \times 2D_\ell}. \tag{2}$$

Stage 3 has no PatchMerging. The resulting spatial side lengths are $S_\ell \in \{64, 32, 16, 8\}$ and the corresponding embedding dimensions are $D_\ell \in \{96, 192, 384, 768\}$ (see Table 2).

Within-stage residual structure. Inside each stage ℓ, both the attention and MLP sub-layers write additively to the residual stream. Following the pre-norm formulation (**?**), the stream after block b is:

$$\mathbf{Z}^{(\ell,b)} = \mathbf{Z}^{(\ell,0)} + \sum_{b'=1}^{b} \mathbf{A}_{\ell,b'} + \sum_{b'=1}^{b} \mathbf{M}_{\ell,b'}, \quad \in \mathbb{R}^{S_\ell^2 \times D_\ell}, \tag{3}$$

where $\mathbf{Z}^{(\ell,0)}$ is the stage input, $\mathbf{A}_{\ell,b}$ is the W-MSA output, $\mathbf{M}_{\ell,b}$ is the MLP output, and S_ℓ is the spatial side length of the token grid at stage ℓ.

This decomposition holds strictly *within* a stage, where D_ℓ is constant. Across stage boundaries, PatchMerging changes both spatial resolution and embedding dimension ($S_\ell \to S_\ell/2$, $D_\ell \to 2D_\ell$), breaking any global additive structure across the full network, unlike in isotropic vision transformers (**?**).

Per-head output of the attention layer. The W-MSA output $\mathbf{A}_{\ell,b}$ is produced by concatenating the outputs of H_ℓ independent attention heads and projecting through a shared output matrix $W^O_{\ell,b}$. At block (ℓ, b), the raw output of head h is (see Appendix A.2 for derivation):

$$\mathbf{H}_{\ell,b,h} = \text{Softmax}\left(\frac{\mathbf{Q}_h \mathbf{K}_h^\top}{\sqrt{d_h}} + \mathbf{B}_{\ell,b,h}\right)\mathbf{V}_h \;\in\; \mathbb{R}^{N_w^\ell \times M \times d_h}, \tag{4}$$

where $N_w^\ell = S_\ell^2/M$ is the number of attention windows, $M = 64$ tokens per window, and $\mathbf{B}_{\ell,b,h} \in \mathbb{R}^{M\times M}$ is the learned relative position bias for head h at block (ℓ, b) (shared across windows, independent per head and per block). All H_ℓ head outputs are concatenated and projected to give the W-MSA output (see Appendix A.3 for the full decomposition):

$$\mathbf{A}_{\ell,b} = \left[\mathbf{H}_{\ell,b,1} \| \cdots \| \mathbf{H}_{\ell,b,H_\ell}\right] W^O_{\ell,b} + \mathbf{b}^O_{\ell,b} \;\in\; \mathbb{R}^{N_w^\ell \cdot M \times D_\ell}. \tag{5}$$

We analyse the raw head outputs $\mathbf{H}_{\ell,b,h}$ directly, *before* the output projection $W^O_{\ell,b}$. Each $\mathbf{H}_{\ell,b,h}$ lives in the per-head space $\mathbb{R}^{d_h}$, which is constant across all stages and blocks. This choice avoids entangling information across heads introduced by the shared projection, and enables uniform cross-stage comparisons of head geometry throughout the network. A complementary analysis of the projected contributions $\widehat{\mathbf{H}}_{\ell,b,h}$—which characterise how each head shapes the residual stream—is provided in Appendix **??**.

Block computation: W-MSA and MLP. Each block b at stage ℓ applies two sub-layers in sequence, both preceded

Table 2. HTS-AT stage-level architectural parameters. The per-head dimension $d_h = D_\ell/H_\ell = 24$ is constant across all stages. S_ℓ is the spatial side length of the token grid at stage ℓ; $N_w^\ell = S_\ell^2/M$ is the number of attention windows, with $M = w^2 = 64$ tokens per window ($w = 8$).

Stage ℓ	**Blocks** B_ℓ	**Heads** H_ℓ	**Dim** D_ℓ	**Spatial** S_ℓ^2	**Windows** N_w^ℓ
0	2	4	96	64×64	64
1	2	8	192	32×32	16
2	6	16	384	16×16	4
3	2	32	768	8×8	1
Total heads H_{tot}	$2{\cdot}4 + 2{\cdot}8 + 6{\cdot}16 + 2{\cdot}32 = 8 + 16 + 96 + 64 = 184$				

by LayerNorm and connected via residual additions:

$$\mathbf{Z}^{(\ell,b)} \leftarrow \mathbf{Z}^{(\ell,b-1)} + \text{W-MSA}_{\ell,b}\Big(\text{LN}\Big(\mathbf{Z}^{(\ell,b-1)}\Big)\Big), \quad (6)$$

$$\mathbf{Z}^{(\ell,b)} \leftarrow \mathbf{Z}^{(\ell,b)} + \text{MLP}_{\ell,b}\Big(\text{LN}\Big(\mathbf{Z}^{(\ell,b)}\Big)\Big). \quad (7)$$

Even-indexed blocks use standard Window Multi-head Self-Attention (W-MSA); odd-indexed blocks use Shifted-Window MSA (SW-MSA), which shifts the partition by $(w/2, w/2)$ tokens to enable cross-window interactions. At Stage 3, where $S_3 = w = 8$ so the grid coincides with a single window, the shift degenerates to zero and all blocks use W-MSA. The MLP sub-layer is a two-layer feed-forward network with hidden dimension $4D_\ell$ and GELU activation.

WindowAttention in detail. The W-MSA module at block (ℓ, b) first partitions the S_ℓ^2 tokens into $N_w^\ell = S_\ell^2/M$ non-overlapping windows of $M = 64$ tokens each, then applies multi-head attention independently within each window.

In the SW-MSA variant, a cyclic shift is applied before window partitioning.

Queries, keys, and values are computed via a single fused projection $W_{\ell,b}^{QKV} \in \mathbb{R}^{D_\ell \times 3D_\ell}$:

$$\text{LN}(\mathbf{Z}^{(\ell,b-1)})\, W_{\ell,b}^{QKV} \xrightarrow{\text{split}(D_\ell)} \mathbf{Q},\, \mathbf{K},\, \mathbf{V} \;\in\; \mathbb{R}^{N_w^\ell \times M \times D_\ell}. \quad (8)$$

Each tensor is then reshaped to isolate the H_ℓ heads, yielding per-head slices $\mathbf{Q}_h, \mathbf{K}_h, \mathbf{V}_h \in \mathbb{R}^{N_w^\ell \times M \times d_h}$ with $d_h = D_\ell/H_\ell = 24$.

For each head $h \in \{1, \ldots, H_\ell\}$, attention is computed as:

$$\mathbf{H}_{\ell,b,h} = \text{Softmax}\left(\frac{\mathbf{Q}_h \mathbf{K}_h^\top}{\sqrt{d_h}} + \mathbf{B}_{\ell,b,h}\right)\mathbf{V}_h \;\in\; \mathbb{R}^{N_w^\ell \times M \times d_h}, \quad (9)$$

where $\mathbf{B}_{\ell,b,h} \in \mathbb{R}^{M \times M}$ is the learned relative position bias for head h of block (ℓ, b). This bias encodes the relative spatial offset between each pair of tokens *within* a window; it is shared across all N_w^ℓ windows (broadcast) and is independent for each head. Concretely, it is read from a parameter table of shape $((2w-1)^2, H_\ell) = (225, H_\ell)$ via a fixed index mapping, so each head h has its own column of learnable values.

All H_ℓ head outputs are then concatenated and passed through the block-specific output projection $W_{\ell,b}^O \in \mathbb{R}^{D_\ell \times D_\ell}$ with bias $\mathbf{b}_{\ell,b}^O \in \mathbb{R}^{D_\ell}$, giving the W-MSA output

$$\mathbf{A}_{\ell,b} = \big[\mathbf{H}_{\ell,b,1} \| \cdots \| \mathbf{H}_{\ell,b,H_\ell}\big]\, W_{\ell,b}^O + \mathbf{b}_{\ell,b}^O \;\in\; \mathbb{R}^{N_w^\ell \cdot M \times D_\ell}, \quad (10)$$

which is added to the residual stream in Eq. (6).

A.3. Per-Head Residual Decomposition

The additive residual structure of Eq. (6) allows us to decompose the W-MSA contribution $\mathbf{A}_{\ell,b}$ exactly into independent per-head terms. Denoting by $W_{\ell,b,h}^O \in \mathbb{R}^{d_h \times D_\ell}$ the row slice of $W_{\ell,b}^O$ corresponding to head h (rows $[(h-1)d_h,\ hd_h)$), we distribute the output projection over heads:

$$\begin{aligned}\mathbf{A}_{\ell,b} &= \big[\mathbf{H}_{\ell,b,1} \| \cdots \| \mathbf{H}_{\ell,b,H_\ell}\big] W_{\ell,b}^O + \mathbf{b}_{\ell,b}^O \\ &= \sum_{h=1}^{H_\ell}\Big(\mathbf{H}_{\ell,b,h}\, W_{\ell,b,h}^O + \tfrac{\mathbf{b}_{\ell,b}^O}{H_\ell}\Big) \;=\; \sum_{h=1}^{H_\ell} \widehat{\mathbf{H}}_{\ell,b,h}, \quad (11)\end{aligned}$$

where we define the *per-head projected contribution*

$$\widehat{\mathbf{H}}_{\ell,b,h} \;=\; \mathbf{H}_{\ell,b,h}\, W_{\ell,b,h}^O + \frac{\mathbf{b}_{\ell,b}^O}{H_\ell} \;\in\; \mathbb{R}^{N_w^\ell \times M \times D_\ell}. \quad (12)$$

This decomposition is exact and follows from the linearity of matrix multiplication: because $W_{\ell,b}^O$ acts on the concatenation of all head outputs, each head h effectively multiplies only its own row slice $W_{\ell,b,h}^O$, and the output bias can be distributed equally among heads without any approximation.

The decomposition holds within a single stage, where D_ℓ is constant. Across stage boundaries, PatchMerging changes both spatial resolution and embedding dimension, breaking any global additive structure; cross-stage comparisons must therefore account for this.

B. Head Analisys

B.1. Spatial Aggregation and Dataset Representations

Each raw head output $\mathbf{H}_{\ell,b,h}$ is a three-dimensional tensor indexed by window, token position, and feature dimension. To obtain a single fixed-size vector per audio sample, we mean-pool over all windows and token positions:

$$\mathbf{r}_{\ell,b,h} = \frac{1}{N_w^\ell \, M} \sum_{i=1}^{N_w^\ell} \sum_{j=1}^{M} \mathbf{H}_{\ell,b,h}[i,j,:] \;\in\; \mathbb{R}^{d_h}. \tag{13}$$

This yields one vector $\mathbf{r}_{\ell,b,h}$ per audio sample, per head h, at every block (ℓ, b) of the network. Stacking these vectors across the n samples in the dataset gives the matrix

$$\mathbf{R}_{\ell,b,h} \in \mathbb{R}^{n \times d_h}, \tag{14}$$

which is the primary object of our analysis. The total number of such matrices across the model is $H_{\text{tot}} = 184$ (one per head per block; see Table 2).

Since $d_h = 24$ is constant across all stages, the ambient dimension of $\mathbf{R}_{\ell,b,h}$ is uniform throughout the network, enabling direct cross-stage comparisons of all dimensionality metrics without correction for a varying ceiling.

B.2. Dataset Representations

Per l'analisi della dimensionalità è stato scelto il dataset esc50 per la sua compattezza e per la sua ricchezza di classi. Infatti ...

B.3. Intrinsic Dimensionality Analysis

To characterize the effective complexity of the projected head representations $\{\mathbf{r}_{\ell,b,h}^{(i)}\}_{i=1}^{n} \subset \mathbb{R}^{1024}$, we employ a battery of linear and nonlinear dimensionality estimators.

B.3.1. Linear Estimators

PCA-based dimensionality. For each head, let $\mathbf{R}_{\ell,b,h} \in \mathbb{R}^{n \times d_h}$ stack all aggregated representations. We compute the covariance matrix $\mathbf{C}_{\ell,b,h} = \frac{1}{n-1}\mathbf{R}^\top \mathbf{R}$ and obtain ordered eigenvalues $\lambda_1 \geq \lambda_2 \geq \cdots$. Linear intrinsic dimensionality is:

$$d_{\text{PCA}}(\alpha) = \arg\min_k \left\{ \frac{\sum_{i=1}^{k} \lambda_i}{\sum_i \lambda_i} \geq \alpha \right\}, \tag{15}$$

evaluated at $\alpha \in \{0.90, 0.95, 0.99\}$. We also report the *Explained Variance Ratio* of the first principal component, $\text{EVR}_1 = \lambda_1 / \sum_i \lambda_i$.

Participation Ratio.

$$\text{PR} = \frac{\left(\sum_i \lambda_i\right)^2}{\sum_i \lambda_i^2}. \tag{16}$$

High PR signals uniform variance distribution; low PR signals dominance of few directions.

Effective Rank.

$$\text{EffRank} = \exp\left(-\sum_i p_i \log p_i \right), \quad p_i = \frac{\lambda_i}{\sum_j \lambda_j}, \tag{17}$$

which can be interpreted as the exponential of the spectral entropy. Unlike PCA thresholds, it provides a smooth, threshold-free notion of dimensionality, capturing how uniformly variance is distributed across components.

B.3.2. Nonlinear Estimators

TwoNN.

$$d_{\text{TwoNN}} = \left(\frac{1}{n} \sum_{i=1}^{n} \log \frac{r_2^{(i)}}{r_1^{(i)}} \right)^{-1}, \tag{18}$$

where $r_1^{(i)}, r_2^{(i)}$ are Euclidean distances to the first and second nearest neighbours of $\mathbf{r}_{\ell,b,h}^{(i)}$ (**?**).

MLE.

$$\hat{d}_{\text{MLE}}(\mathbf{r}) = \left(\frac{1}{k-1} \sum_{j=1}^{k-1} \log \frac{r_k(\mathbf{r})}{r_j(\mathbf{r})} \right)^{-1}, \tag{19}$$

averaged over all samples (**?**), with $k = 20$. This estimator is based on a local maximum likelihood principle, assuming uniform sampling in small neighbourhoods, and is sensitive to fine-scale manifold structure beyond global linear approximations.

Linear-Nonlinear (L/N) Ratio

$$\text{Ratio}_B = \frac{\bar{d}_{\text{PCA}_{99}}}{\bar{d}_{\text{TwoNN}}}. \tag{20}$$

Values near 1 indicate near-linear manifolds; higher values signal nonlinear curvature beyond what PCA captures, and serve as a diagnostic for selecting layer targets for spectral reweighting.

B.4. Results: Dimensionality Evolution Across the Hierarchy

L'analisi della dimensionalità rivela una transizione strutturata nel modo in cui l'informazione acustica viene elaborata attraverso i blocchi di HTSAT. I risultati aggregati per blocco sono riportati in Figura 2, mentre il dettaglio per singola testa è illustrato in Figura 3.

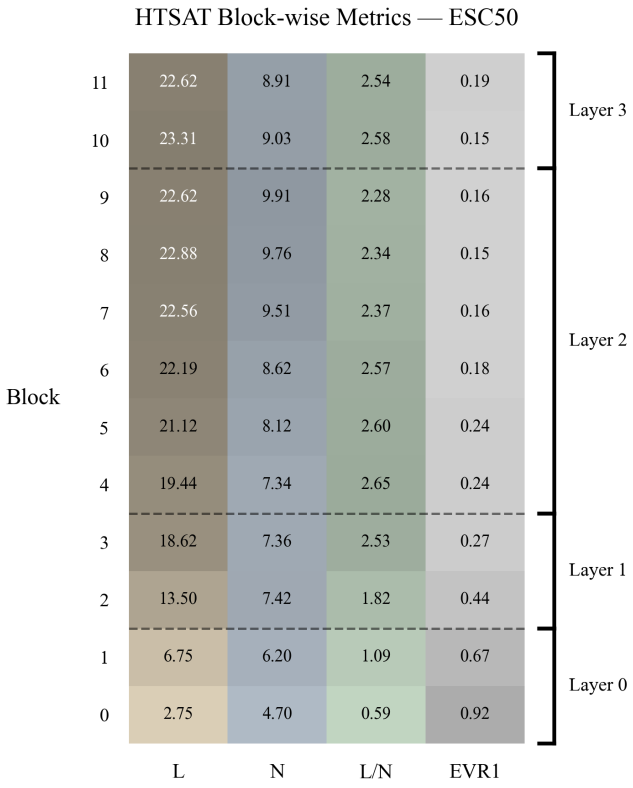


Figure 2. **Block-wise Metrics per il dataset ESC50.** La heatmap mostra l'andamento medio per ogni blocco (0-11) della dimensionalità lineare (L), nonlineare (N), del rapporto L/N e della dominanza della prima componente principale (EVR1). Le linee tratteggiate separano i quattro stage (Layer 0-3) dell'architettura.

B.4.1. Global Trends and Stage-wise Progression

Osservando la Figura 2, emerge chiaramente come la complessità delle rappresentazioni aumenti progressivamente risalendo la gerarchia del modello.

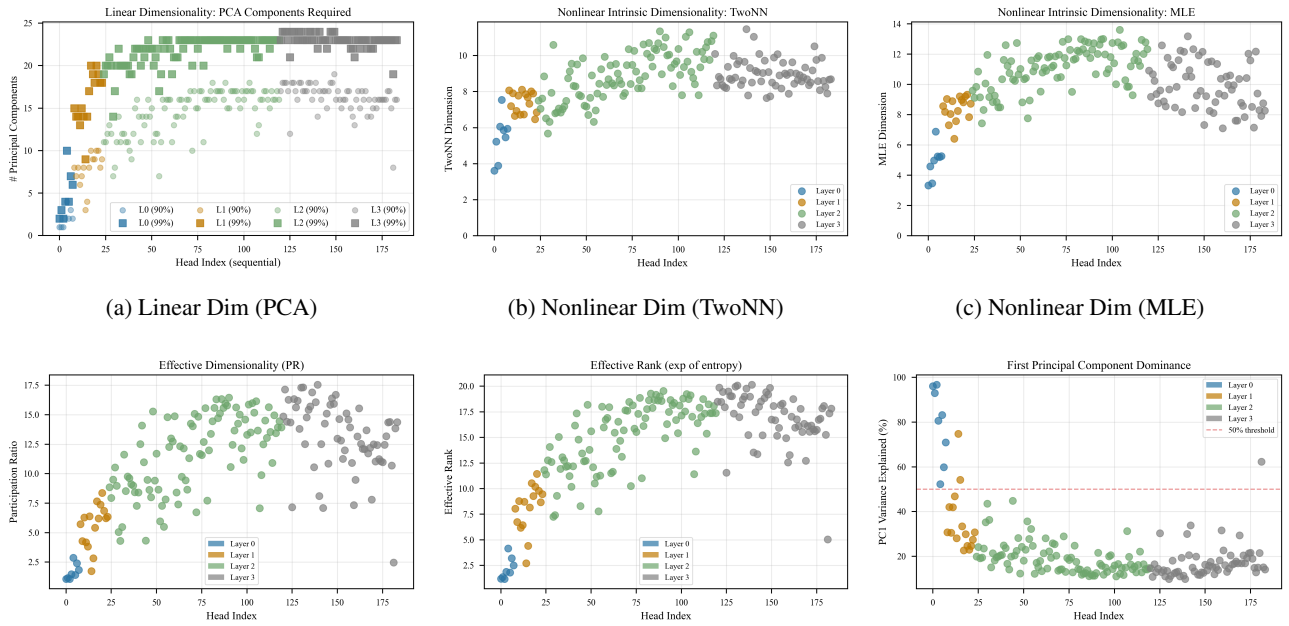
- **Layer 0 (Expansion Phase):** Nei primi blocchi, la dimensionalità intrinseca è minima ($L \approx 2.75$ nel blocco 0). L'altissimo valore di $\mathrm{EVR1}$ (0.92) indica che quasi tutta la varianza è concentrata in una singola direzione. Questo suggerisce che i primi stadi agiscono come estrattori di feature a bassa complessità, focalizzandosi su segnali dominanti del dominio tempo-frequenza.
- **Layer 1 & 2 (Complexity Growth):** Si osserva una crescita rapida di L e N. Il rapporto L/N raggiunge il suo picco nel Layer 2 (2.65), suggerendo che in questa fase la curvatura del manifold è massima. Le rappresentazioni non solo occupano più dimensioni, ma lo fanno in modo significativamente non lineare rispetto alle basi PCA.
- **Layer 3 (Saturation and Refinement):** Negli ultimi blocchi (10-11), la dimensionalità tende a stabilizzarsi o a subire una leggera contrazione (specialmente in N). La $\mathrm{EVR1}$ crolla ai minimi storici (0.15), indicando una distribuzione della varianza estremamente uniforme tra le componenti disponibili ($d_h = 24$).

B.4.2. Head-specific Analysis and Manifold Structure

L'analisi a livello di singola testa (Figura 3) permette di apprezzare l'eterogeneità funzionale all'interno dello stesso blocco.

I grafici evidenziano alcuni fenomeni chiave:

- **Saturation of Linear Capacity:** Il panel (A) mostra che molte teste nei Layer 2 e 3 raggiungono la dimensionalità massima consentita dalla soglia $\alpha = 99\%$, saturando vicino al limite dell'ambient dimension ($d_h = 24$).
- **Effective Rank vs. PR:** Sia l'Effective Rank (E) che il Participation Ratio (D) confermano una transizione da un regime a "basso rango" (Layer 0) a un regime ad "alto rango" nei layer profondi. Tuttavia, la dispersione nel Layer 2 suggerisce che alcune teste mantengano una specializzazione più focalizzata (bassa dimensionalità) rispetto ad altre che esplorano lo spazio intero.
- **PC1 Dominance Decay:** Il panel (F) illustra la "decostruzione" della dominanza spettrale. Mentre le



(a) Linear Dim (PCA) (b) Nonlinear Dim (TwoNN) (c) Nonlinear Dim (MLE)

(d) Participation Ratio (e) Effective Rank (f) PC1 Dominance

Figure 3. **Analisi granulare delle 184 teste di HTSAT.** I panel mostrano la distribuzione delle metriche di dimensionalità per ogni testa, indicizzate sequenzialmente. (A-C) Confronto tra stime lineari e non lineari; (D-E) Metriche di diffusione spettrale; (F) Concentrazione della varianza sulla prima componente.

prime teste (Layer 0) hanno una PC1 che spiega quasi il 100% della varianza, nelle teste finali tale valore scende sotto il 20%, un segnale di rappresentazioni distribuite e potenzialmente più robuste.

Reflections on Representational Complexity. Questa evoluzione suggerisce che HTSAT operi una progressiva *unfolding* dei dati audio: da proiezioni iniziali quasi unidimensionali (probabilmente legate a feature energetiche o spettrali grezze) verso manifold complessi e multidimensionali nei layer intermedi, dove avviene la discriminazione semantica delle classi di ESC50. L'alto rapporto L/N nei Layer 2 e 3 identifica questi blocchi come i candidati ideali per interventi di regolarizzazione spettrale, essendo i punti di massima compressione non lineare dell'informazione.

B.5. Head Specialization Analysis

Per interpretare il contenuto semantico codificato dalle singole teste, introduciamo un framework di *cross-modal grounding* basato su un modello CLAP pre-addestrato. L'obiettivo è mappare l'asse di varianza principale di ogni testa in uno spazio di concetti testuali descrittivi.

B.5.1. Semantic Mapping via Principal Component Projection

Sia $\mathbf{R}_{\ell,b,h} \in \mathbb{R}^{n \times d_h}$ la matrice delle rappresentazioni aggregate per una data testa. Per identificare la caratteristica acustica primaria catturata dalla testa, calcoliamo la prima componente principale $\mathbf{v}_1 \in \mathbb{R}^{d_h}$. Per ogni campione i, lo score sulla prima componente è definito come:

$$s_i = \mathbf{r}_{\ell,b,h}^{(i)} \cdot \mathbf{v}_1. \tag{21}$$

Definiamo quindi la **Direzione Semantica** $\sigma_{\ell,b,h}$ nello spazio latente CLAP come la combinazione lineare degli embedding audio CLAP $\{\mathbf{a}_i\}_{i=1}^n \subset \mathbb{R}^{D_{\text{CLAP}}}$, pesata dagli score normalizzati:

$$\sigma_{\ell,b,h} = \sum_{i=1}^{n} w_i \mathbf{a}_i, \quad \text{con } w_i = \frac{s_i - \bar{s}}{\|\mathbf{s} - \bar{s}\|_2}, \tag{22}$$

dove $\bar{s}$ è la media degli score. Questo vettore σ rappresenta l'asse di massima variazione della testa proiettato in uno spazio allineato al linguaggio naturale.

B.5.2. Sparse Coding via Orthogonal Matching Pursuit (OMP)

Per tradurre $\sigma_{\ell,b,h}$ in termini comprensibili, utilizziamo un dizionario testuale $\mathcal{D}$ composto da M descrizioni audio (es. "dog barking", "high pitched tone"). Siano $\{\mathbf{t}_j\}_{j=1}^M$ i

corrispondenti embedding testuali CLAP. Poiché una testa potrebbe rispondere a una combinazione di concetti, decomponiamo σ mediante un algoritmo *Orthogonal Matching Pursuit* (OMP). L'OMP seleziona iterativamente i k atomi dal dizionario che meglio spiegano la direzione semantica, risolvendo:

$$\min_{\mathbf{x}} \|\sigma_{\ell,b,h} - \mathbf{T}\mathbf{x}\|_2 \quad \text{s.t. } \|\mathbf{x}\|_0 \leq k, \tag{23}$$

dove $\mathbf{T}$ è la matrice degli embedding del dizionario. Nel nostro esperimento fissiamo $k = 5$. Questo approccio "greedy" permette di distinguere tra teste altamente specializzate (che puntano a un singolo atomo) e teste che catturano concetti acustici più ampi.

B.5.3. Metrics for Semantic Coherence

Per quantificare la qualità della specializzazione, introduciamo lo **Z-score di Coerenza**. Data la sottorete di k descrizioni selezionate $\mathcal{D}_k \subset \mathcal{D}$, definiamo la similarità intra-gruppo S_{intra} come la media dei prodotti scalari tra gli embedding selezionati. Lo Z-score è calcolato come:

$$Z = \frac{S_{\text{intra}} - \mu_{\mathcal{D}}}{\sigma_{\mathcal{D}}}, \tag{24}$$

dove $\mu_{\mathcal{D}}$ e $\sigma_{\mathcal{D}}$ sono rispettivamente la media e la deviazione standard delle similarità tra gli atomi selezionati e l'intero dizionario $\mathcal{D}$. Un valore di Z elevato indica che la testa è focalizzata su un dominio semantico ben definito e consistente (es. solo suoni naturali o solo parametri timbrici), mentre valori bassi indicano una specializzazione spuria o eccessivamente eterogenea.

B.6. Results: Semantic Landscapes and Head Specialization

L'applicazione del framework di *cross-modal grounding* rivela l'emergere di una specializzazione funzionale che segue la gerarchia strutturale del modello. L'analisi dello Z-score di coerenza semantica (Figura 4) e l'identificazione delle teste più specializzate (Tabella 4) permettono di capire come i concetti acustici astratti si cristallizzino in categorie semantiche.

B.6.1. Emergence of Semantic Coherence

L'andamento della coerenza semantica attraverso i layer (Figura 4) rivela una transizione di fase legata alla struttura gerarchica del modello:

- **Layer 0 e 1 (Semantic Void):** Nei primi quattro blocchi (0-3), la coerenza semantica è prevalentemente negativa o prossima allo zero. In questa fase, le teste estraggono feature a bassa dimensionalità ($L < 14$) e altissima dominanza spettrale ($\text{EVR}_1 \approx 0.92$). La bassa coerenza i ndica che queste teste rispondono a proprietà fisiche del segnale comuni a molteplici sorgenti, rendendo impossibile un mapping univoco su descrizioni testuali specifiche.

- **Layer 2 (Functional Diversification):** This stage marks the transition into a regime of increased non-linearity. In particular, a clear increase in head specialization emerges toward the end of the layer, in concomitance with the growth of the intrinsic non-linear dimensionality (as measured by TwoNN, Fig. 2) and the concurrent decrease in the leading eigenvalue share (EVR_1), which drops from 0.24 at the beginning of Layer 2 to 0.16 at its end (Fig. 2).

  This joint behaviour suggests that the expansion of the underlying manifold complexity is closely linked to the emergence of more semantically selective attention heads. Accordingly, highly specialized heads begin to appear, such as `L2_B4_H1` ($Z = 1.46$, absolute block 8), which capture fine-grained acoustic events associated with the *human_body* category.

- **Layer 3 (Semantic Convergence):** This layer initially continues the trend observed at the end of Layer 2, with a high density of more specialized heads in the early blocks of Layer 3. In this regime, heads remain well-aligned with CLAP semantic concepts, despite the fact that the leading eigenvalue share (EVR_1) reaches one of its lowest values (0.15), indicating that information is distributed across a larger number of latent components.

  However, this trend progressively reverses toward the final blocks (10–11), where a mild decrease in head specialization is observed. This reversal is consistent with a broader inversion in all structural metrics, including L, N, L/N, and EVR_1, which collectively shift from their previously increasing or decreasing trends (see Fig. 2).

  Overall, Layer 3 appears to exhibit an initial phase of strong semantic alignment, followed by a late-stage relaxation of specialization, suggesting a partial remixing of representations in the deepest blocks.

B.6.2. Identification of Specialized Heads

La Tabella 4 riporta una selezione delle teste che mostrano la massima coerenza semantica, evidenziando una predilezione per categorie con forti connotati timbrici o temporali.

The results reported in Table 4 confirm the emergence of a clear semantic organization of attention heads across layers. In particular, the highest coherence scores are consistently associated with heads that capture well-defined

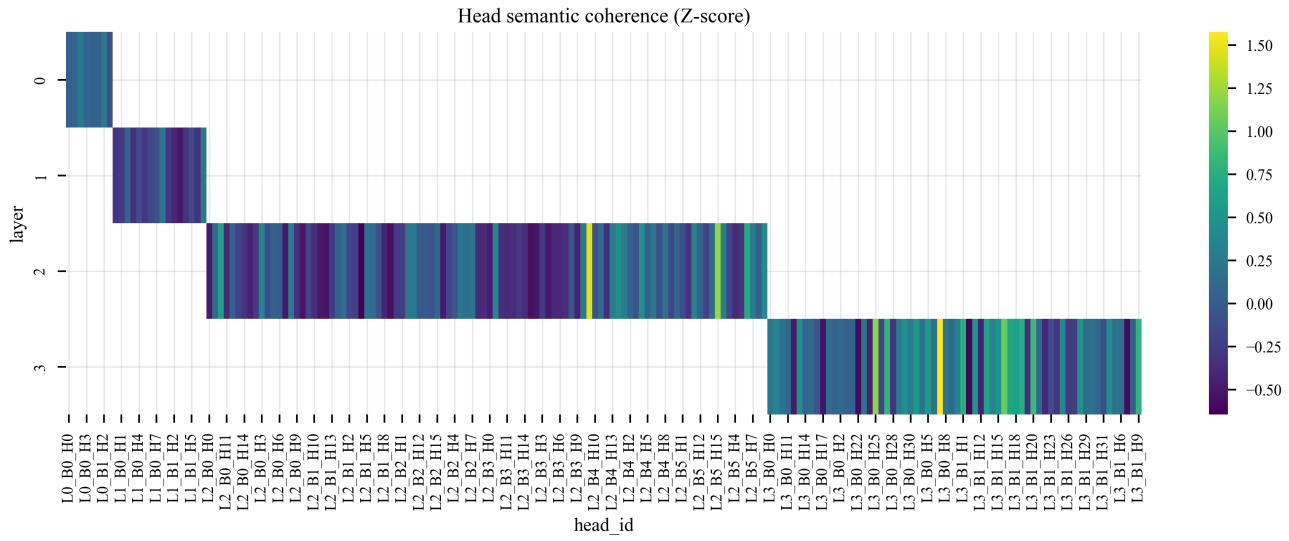


Figure 4. **Heatmap dello Z-score di coerenza semantica.** La mappa evidenzia come la capacità di allineamento semantico emerga drasticamente a partire dalla fine del Layer 2, in corrispondenza dell'aumento della dimensionalità non-lineare e della riduzione della dominanza della prima componente principale.

acoustic categories such as *human_body*, *nature*, and *urban sounds*, suggesting that specialization is not random but structurally aligned with meaningful audio concepts.

A notable trend is the predominance of highly coherent heads in Layers 2 and 3, supporting the observation that semantic specialization intensifies after the initial low-level feature extraction stage. Moreover, heads with higher Z-scores tend to exhibit lower PC1 variance, indicating that stronger semantic alignment correlates with more concentrated spectral structure.

Finally, the presence of moderately coherent but still structured heads in later blocks suggests that, while full specialization emerges primarily in intermediate layers, the deepest representations undergo a partial re-mixing of features, consistent with the non-monotonic behaviour observed in Fig. 4 and Fig. 2. Importantly, the observed inversion of the trend in the deepest blocks is also consistent with the findings reported in (**?**), indicating a shared structural behaviour across architectures and suggesting that this non-monotonic specialization pattern extends beyond the audio domain to image-domain residual stream models as well.

C. Reweighting Pipeline

C.1. ResiDual: Implementation Details

This section describes the full implementation of ResiDual (RD*), a learnable spectral adaptation integrated into a frozen CLAP model. The pipeline consists of two sequential phases: a *fitting pass* that estimates a fixed PCA basis for each attention head (§C.1.1), followed by an *optimisation phase* in which the per-head weight vectors are refined as free parameters via gradient descent to maximise zero-shot classification accuracy (§C.1.3). The forward-pass reweighting applied at inference time is described in §C.1.2.

Throughout this appendix we assume batch size $B = 1$ for notational clarity; all operations extend straightforwardly to $B > 1$ by treating the batch axis independently. We use the index conventions of Appendix A: $\ell \in \{0, 1, 2, 3\}$ is the stage index, $b \in \{1, \dots, B_\ell\}$ is the block index within stage ℓ, and $h \in \{1, \dots, H_\ell\}$ is the head index. The per-head feature dimension is $d_h = 24$ (constant across all stages), $N_w^\ell = S_\ell^2/M$ is the number of attention windows at stage ℓ, and $M = 64$ is the number of tokens per window. All symbols are collected in Table 3.

Spectral reweighting layers. For each target stage $\ell \in \mathcal{L}$, block b, and head h, RD*instantiates a `SpectralReweightingLayer` — a lightweight `nn.Module` that carries two non-trainable buffers, $\mathbf{\Phi}_{\ell,b,h} \in \mathbb{R}^{d_h \times d_h}$ and $\boldsymbol{\mu}_{\ell,b,h} \in \mathbb{R}^{d_h}$, and a single learnable parameter vector $\boldsymbol{\lambda}_{\ell,b,h} \in \mathbb{R}^{d_h}$. All such modules are collected in a nested `nn.ModuleDict` indexed by (ℓ, b, h), constructed once during model initialisation in `_build_spectral_layers` from the stage dimensions and head counts of the underlying HTS-AT configuration; they persist in the model's `state_dict` across saves and loads. Each module exposes two methods: `fit_pca`, which populates $\mathbf{\Phi}_{\ell,b,h}$ and $\boldsymbol{\mu}_{\ell,b,h}$ from data and initialises $\boldsymbol{\lambda}_{\ell,b,h}$ (§C.1.1); and `forward`, which applies the RD*transform (§C.1.2).

Module injection via forward hooks. RD*injects its reweighting through PyTorch forward hooks. For each target block (ℓ, b) with $\ell \in \mathcal{L}$, a dedicated `AttentionHook` object is registered on the corresponding `WindowAttention` module immediately before the forward pass begins and removed as soon as that forward pass completes. The hook is stateless: it holds only a reference to the corresponding slice `spectral_layers`$[\ell][b]$ of the module dict, and operates in one of two modes controlled by a `collect_for_fitting` flag. In *collection mode*, it accumulates the raw per-head tensors $\mathbf{H}_{\ell,b,h}$ without modifying the forward-pass output, feeding the fitting pass of §C.1.1. In *reweighting mode*, it calls `forward` on each `SpectralReweightingLayer` and replaces $\mathbf{H}_{\ell,b,h}$ with the reweighted output $\mathbf{H}^*_{\ell,b,h}$ before concatenation and output projection. No structural modification of the original model is performed: all pretrained parameters — the fused QKV projection $W^{QKV}_{\ell,b}$, the output projection $W^{O}_{\ell,b}$ with bias $\mathbf{b}^{O}_{\ell,b}$, and the relative position bias table $\mathbf{B}_{\ell,b,h}$ — remain exactly as loaded from the checkpoint.

C.1.1. Fitting Pass: PCA Basis Estimation

Data collection. The fitting pass is run with gradients disabled and the model in evaluation mode. For each audio clip (up to $N_{\max}$ clips), a forward pass is executed with data collection enabled. The hook registered at block (ℓ, b) intercepts, for every head h, the tensor $\mathbf{H}_{\ell,b,h} \in \mathbb{R}^{N^{\ell}_{w} \times M \times d_h}$ and flattens it along the spatial axes to obtain a matrix of shape $\mathbb{R}^{(N^{\ell}_{w} \cdot M) \times d_h}$, which is appended to an accumulation buffer for the triplet (ℓ, b, h). After all clips have been processed, the per-clip contributions are stacked to form the *fitting matrix*

$$\widetilde{\mathbf{R}}_{\ell,b,h} \in \mathbb{R}^{N_{\text{fit}} \times d_h}, \qquad N_{\text{fit}} = N_{\text{clips}} \cdot N^{\ell}_{w} \cdot M, \quad (25)$$

where $N_{\text{clips}} \leq N_{\max}$ is the number of processed clips.

PCA decomposition. Standard PCA is applied to $\widetilde{\mathbf{R}}_{\ell,b,h}$. The empirical mean

$$\boldsymbol{\mu}_{\ell,b,h} = \frac{1}{N_{\text{fit}}} \sum_{i=1}^{N_{\text{fit}}} \widetilde{\mathbf{R}}_{\ell,b,h}[i,:] \in \mathbb{R}^{d_h} \quad (26)$$

is computed and subtracted, and the centred empirical covariance is decomposed as

$$\begin{aligned} \mathbf{C}_{\ell,b,h} &= \frac{1}{N_{\text{fit}} - 1} \left(\widetilde{\mathbf{R}}_{\ell,b,h} - \mathbf{1}\boldsymbol{\mu}^{\top}_{\ell,b,h} \right)^{\top} \left(\widetilde{\mathbf{R}}_{\ell,b,h} - \mathbf{1}\boldsymbol{\mu}^{\top}_{\ell,b,h} \right) \\ &= \boldsymbol{\Phi}_{\ell,b,h} \, \operatorname{diag}(\lambda^{\text{ev}}_1, \ldots, \lambda^{\text{ev}}_{d_h}) \, \boldsymbol{\Phi}^{\top}_{\ell,b,h}, \quad \lambda^{\text{ev}}_1 \geq \cdots \geq \lambda^{\text{ev}}_{d_h} \geq 0, \end{aligned} \quad (27)$$

where $\boldsymbol{\Phi}_{\ell,b,h} \in \mathbb{R}^{d_h \times d_h}$ is the orthogonal matrix of eigenvectors (columns ordered by decreasing eigenvalue) and λ^{ev}_j denotes the j-th eigenvalue of $\mathbf{C}_{\ell,b,h}$. We use the superscript "ev" throughout to distinguish these fixed quantities from the trainable weight vector $\boldsymbol{\lambda}_{\ell,b,h}$. Both $\boldsymbol{\Phi}_{\ell,b,h}$ and $\boldsymbol{\mu}_{\ell,b,h}$ are stored as non-trainable buffers; the accumulation buffers $\widetilde{\mathbf{R}}_{\ell,b,h}$ are then cleared to free memory.

Task-relevant subspace and initialisation of $\boldsymbol{\lambda}_{\ell,b,h}$. For each head (ℓ, b, h), the index $k_{\ell,b,h}$ defined in Eq. (**??**) identifies the smallest number of principal components that together explain at least a fraction ρ_{var} of the total variance, providing a per-head estimate of the effective dimensionality of the task-relevant subspace. The learnable weight vector $\boldsymbol{\lambda}_{\ell,b,h} \in \mathbb{R}^{d_h}$ is initialised component-wise as

$$\lambda^{(0)}_j = \begin{cases} 1 & j \leq k_{\ell,b,h}, \\ 0 & j > k_{\ell,b,h}, \end{cases} \quad (28)$$

corresponding to the identity map restricted to the task-relevant subspace and complete suppression of the remaining noise components. At this initialisation point, RD*therefore acts as a projection-and-reconstruction operator that retains the top-$k_{\ell,b,h}$ PCA directions unchanged and zeros out the remainder.

C.1.2. Forward Pass with Spectral Reweighting

At inference time, the hook at block (ℓ, b) intercepts the raw head outputs $\mathbf{H}_{\ell,b,h} \in \mathbb{R}^{N^{\ell}_{w} \times M \times d_h}$ (cf. Eq. (4)) and applies the following sequence of operations independently for each head h.

Step 1 — Centering.

$$\widetilde{\mathbf{H}}_{\ell,b,h} = \mathbf{H}_{\ell,b,h} - \boldsymbol{\mu}_{\ell,b,h} \in \mathbb{R}^{N^{\ell}_{w} \times M \times d_h}, \quad (29)$$

where $\boldsymbol{\mu}_{\ell,b,h} \in \mathbb{R}^{d_h}$ is broadcast over the first two axes.

Step 2 — Projection onto the full PCA basis.

$$\mathbf{Z}_{\ell,b,h} = \widetilde{\mathbf{H}}_{\ell,b,h} \, \boldsymbol{\Phi}_{\ell,b,h} \in \mathbb{R}^{N^{\ell}_{w} \times M \times d_h}. \quad (30)$$

Unlike fixed-weight variants that truncate to $k_{\ell,b,h} < d_h$ components, RD*projects onto the *full* basis $\boldsymbol{\Phi}_{\ell,b,h} \in \mathbb{R}^{d_h \times d_h}$, delegating the effective truncation to the learned weights.

Step 3 — Per-component rescaling.

$$\widehat{\mathbf{Z}}_{\ell,b,h} = \mathbf{Z}_{\ell,b,h} \odot \boldsymbol{\lambda}^{\top}_{\ell,b,h} \in \mathbb{R}^{N^{\ell}_{w} \times M \times d_h}, \quad (31)$$

where $\boldsymbol{\lambda}^{\top}_{\ell,b,h} \in \mathbb{R}^{d_h}$ is broadcast over the first two axes. Because $\boldsymbol{\lambda}_{\ell,b,h}$ is a free real-valued parameter, each entry λ_j can amplify ($|\lambda_j| > 1$), attenuate ($|\lambda_j| \in (0, 1)$), suppress ($\lambda_j = 0$), or invert ($\lambda_j < 0$) the corresponding principal component.

Step 4 — Reconstruction in the original space.

$$\mathbf{H}^*_{\ell,b,h} = \widehat{\mathbf{Z}}_{\ell,b,h} \, \boldsymbol{\Phi}^{\top}_{\ell,b,h} \in \mathbb{R}^{N^{\ell}_{w} \times M \times d_h}. \quad (32)$$

The empirical mean $\boldsymbol{\mu}_{\ell,b,h}$ is *not* restored after reconstruction. Since $\mathbf{H}^*_{\ell,b,h}$ is immediately passed to the output projection $W^O_{\ell,b}$ (cf. Eq. (10)), which carries its own learnable bias $\mathbf{b}^O_{\ell,b} \in \mathbb{R}^{D_\ell}$, re-adding $\boldsymbol{\mu}_{\ell,b,h}$ would introduce a systematic, data-dependent shift in the residual stream that is already absorbed by $\mathbf{b}^O_{\ell,b}$.

Combining Steps 1–4, the full RD*transform reads:

$$\mathrm{RD}^*(\mathbf{H}_{\ell,b,h},\, \boldsymbol{\lambda}_{\ell,b,h}) \;=\; \big(\mathbf{H}_{\ell,b,h} - \boldsymbol{\mu}_{\ell,b,h}\big)\, \boldsymbol{\Phi}_{\ell,b,h}\, \mathrm{diag}(\boldsymbol{\lambda}_{\ell,b,h})\, \boldsymbol{\Phi}_{\ell,b,h}^\top \in \mathbb{R}^{N_w \times M \times d_h} \tag{33}$$

The tensor $\mathbf{H}^*_{\ell,b,h}$ replaces $\mathbf{H}_{\ell,b,h}$ in Eq. (10), so the output projection $W^O_{\ell,b}$ and the residual-stream update of Eq. (6) proceed unchanged. During optimisation, gradients of any downstream loss flow through RD* and update only $\boldsymbol{\lambda}_{\ell,b,h}$; the basis $\boldsymbol{\Phi}_{\ell,b,h}$, the mean $\boldsymbol{\mu}_{\ell,b,h}$, and all original model parameters are held fixed.

C.1.3. Optimisation of $\boldsymbol{\lambda}$ via Zero-Shot Accuracy

After the fitting pass has fixed the PCA bases $\{\boldsymbol{\Phi}_{\ell,b,h}\}$ and initialised $\{\boldsymbol{\lambda}_{\ell,b,h}\}$ via Eq. (28), the second phase optimises the weight vectors to maximise zero-shot classification accuracy on a held-out validation set.

Parameter isolation. All model parameters except the weight vectors $\{\boldsymbol{\lambda}_{\ell,b,h}\}_{\ell\in\mathcal{L},\, b,\, h}$ are frozen by setting `requires_grad = False`. The only degrees of freedom during optimisation are therefore the $\sum_{\ell\in\mathcal{L}} B_\ell \cdot H_\ell$ weight vectors, each of dimension $d_h = 24$, that control the per-component gain of each attention head in the target stages $\mathcal{L}$. The CLAP audio projection head $P : \mathbb{R}^{D_3} \to \mathbb{R}^{d_{\text{proj}}}$ is kept frozen but is included in the computational graph so that gradients flow from the similarity space back to $\boldsymbol{\lambda}_{\ell,b,h}$.

Objective. Given a set of audio clips with ground-truth class labels $y \in \{1, \ldots, C\}$ and pre-computed ℓ_2-normalised text embeddings $\mathbf{t}_c \in \mathbb{R}^{d_{\text{proj}}}$ for each class c, the optimisation objective is the cross-entropy loss over zero-shot logits:

$$\mathcal{J}(\boldsymbol{\lambda}) \;=\; -\,\mathbb{E}_{(x,\,y)}\left[\log \frac{\exp\big(\hat{\mathbf{a}}(x)^\top \mathbf{t}_y\big)}{\sum_{c=1}^{C} \exp\big(\hat{\mathbf{a}}(x)^\top \mathbf{t}_c\big)}\right], \tag{34}$$

where $\hat{\mathbf{a}}(x) \in \mathbb{R}^{d_{\text{proj}}}$ is the ℓ_2-normalised audio embedding produced by P applied to the spatially averaged Stage-3 output of HTS-AT after the RD*transforms have been applied (cf. Table 3). Minimising $\mathcal{J}$ is equivalent to maximising zero-shot classification accuracy in expectation.

Optimiser and schedule. The weight vectors $\{\boldsymbol{\lambda}_{\ell,b,h}\}$ are updated with Schedule-Free Adam (**?**) at learning rate η. This optimiser eliminates the need for a separate learning-rate schedule while matching or exceeding the performance of Adam with cosine decay on this class of fine-tuning tasks. When the `schedulefree` package is unavailable, standard Adam is used as a fallback.

Early stopping and best-$\boldsymbol{\lambda}$ restoration. At the end of each epoch, zero-shot accuracy on the validation set is evaluated. A snapshot of the current $\{\boldsymbol{\lambda}_{\ell,b,h}\}$ is saved whenever a new accuracy maximum is achieved. Training halts early if no improvement is observed for p consecutive epochs; at termination, the best snapshot is restored, guaranteeing that the final weight vectors correspond to the configuration achieving the highest observed validation accuracy.

Hyperparameter summary. Table 5 collects all hyperparameters introduced by RD*together with the default values used in all experiments.

Remark on hook-based extraction. The `AttentionHook` mechanism described in §B.1 for the dimensionality analysis is the same mechanism through which RD*applies reweighting at inference time. The hook operates in two distinct modes controlled by an internal `collect_for_fitting` flag: in *collection mode* it accumulates raw head outputs $\mathbf{H}_{\ell,b,h}$ for PCA estimation without modifying the forward-pass output; in *reweighting mode* it replaces $\mathbf{H}_{\ell,b,h}$ with $\mathbf{H}^*_{\ell,b,h}$ (Eq. (32)) before the concatenation and output projection of Eq. (10). In both modes the hooks are *temporary*: they are installed at the start of each call to the instrumented forward function and removed immediately upon its completion, so the underlying `WindowAttention` modules are never structurally modified.

D. Appendice Ultima

Approximation scope and token-level reweighting. The mean-pooled representation $\mathbf{r}_{\ell,b,h}$ is used exclusively for the dimensionality analysis reported in this section. Pooling over all $N^\ell_w \cdot M$ spatial positions collapses the intra-sample token variance, retaining only the inter-sample variance that drives the PCA basis estimation. This choice is motivated by computational constraints: constructing the full token-level matrix $\widetilde{\mathbf{R}}_{\ell,b,h} \in \mathbb{R}^{n\, N^\ell_w M \times d_h}$ for all 184 heads simultaneously would exceed available memory (considering the all linear and non linear estimators). Despite this approximation, the pooled statistics provide a faithful characterisation of the dominant geometric structure of each head's subspace, since the inter-sample variance captures the directions along which heads discriminate between audio inputs—which is precisely the quantity

relevant to understanding head specialisation.

The spectral reweighting in ResiDual, as specified in the next sections, operates instead at full spatial resolution.This token-level granularity allows ResiDual to modulate the geometry of individual spatial positions rather than acting on a global summary, making the intervention more targeted without sacrificing the precision of the basis estimation.

Hook-based extraction. To obtain $\mathbf{H}_{\ell,b,h}$ without modifying the model, we register forward hooks on the `WindowAttention` module of each block (ℓ, b). The hook intercepts the tensor `(attn @ v)` immediately after the per-head attention computation of Eq. (4), yielding the raw outputs $\mathbf{H}_{\ell,b,h}$ of shape $(N_w^\ell,\ M,\ d_h)$. No further projection is applied: the hook output is spatially pooled directly via Eq. (13) to produce $\mathbf{r}_{\ell,b,h}$.

Dataset sampling. We extract representations from four audio classification benchmarks:

- **ESC-50** (**?**): 50 environmental sound classes, 2,000 clips (5 s, 44.1 kHz).
- **TinySOL** (**?**): 14 orchestral instrument classes with varied articulations, 2,071 monophonic samples (1–16 s, 44.1 kHz).
- **VocalSound** (**?**): 6 non-speech vocal categories, stratified subset of 1,200 samples.
- **IRMAS** (**?**): 11 musical instrument classes, annotated audio excerpts with predominant instrument.

Audio preprocessing follows the CLAP standard pipeline: 64-band log-mel spectrogram ($f_{\min} = 50\,\text{Hz}$, $f_{\max} = 8000\,\text{Hz}$, FFT window 1024, hop 320), padded or truncated to 7 seconds at 44.1 kHz.

Table 3. Summary of notation used throughout the paper. Symbols introduced by RD* are marked with $(\star)$.

Symbol	**Definition**
$\ell \in \{0, 1, 2, 3\}$	Stage index
$b \in \{1, \ldots, B_\ell\}$	Block index within stage ℓ
$h \in \{1, \ldots, H_\ell\}$	Attention head index within block (ℓ, b)
S_ℓ	Spatial side length of the token grid at stage ℓ
$w = 8$	Attention window side length
$M = w^2 = 64$	Tokens per attention window
$N_w^\ell = S_\ell^2 / M$	Number of attention windows at stage ℓ
$d_h = 24$	Per-head feature dimension (constant across stages)
$D_\ell = H_\ell \cdot d_h$	Total embedding dimension at stage ℓ
$d_{\text{proj}} = 1024$	CLAP joint embedding dimension
$W_{\ell,b}^{QKV} \in \mathbb{R}^{D_\ell \times 3D_\ell}$	Fused QKV projection at block (ℓ, b)
$W_{\ell,b}^{O} \in \mathbb{R}^{D_\ell \times D_\ell}$	Output projection at block (ℓ, b)
$W_{\ell,b,h}^{O} \in \mathbb{R}^{d_h \times D_\ell}$	Row slice of $W_{\ell,b}^{O}$ for head h
$\mathbf{b}_{\ell,b}^{O} \in \mathbb{R}^{D_\ell}$	Output projection bias at block (ℓ, b)
$\mathbf{B}_{\ell,b,h} \in \mathbb{R}^{M \times M}$	Relative position bias for head h at block (ℓ, b)
$\mathbf{H}_{\ell,b,h} \in \mathbb{R}^{N_w^\ell \times M \times d_h}$	Raw head output at block (ℓ, b), head h (pre-$W_{\ell,b}^{O}$)
$\widehat{\mathbf{H}}_{\ell,b,h} \in \mathbb{R}^{N_w^\ell \times M \times D_\ell}$	Per-head projected contribution (post-$W_{\ell,b}^{O}$); see Eq. (12)
$\mathbf{r}_{\ell,b,h} \in \mathbb{R}^{d_h}$	Spatially aggregated raw head output; see Eq. (13)
$\widehat{\mathbf{r}}_{\ell,b,h} \in \mathbb{R}^{D_\ell}$	Spatially aggregated projected head output; see Eq. (**??**)
$\mathbf{R}_{\ell,b,h} \in \mathbb{R}^{n \times d_h}$	Dataset matrix of $\mathbf{r}_{\ell,b,h}$ across n samples
$\widehat{\mathbf{R}}_{\ell,b,h} \in \mathbb{R}^{n \times D_\ell}$	Dataset matrix of $\widehat{\mathbf{r}}_{\ell,b,h}$ across n samples
$P : \mathbb{R}^{768} \to \mathbb{R}^{1024}$	CLAP audio projection head (two-layer MLP with residual)
n	Number of audio samples in the dataset
Symbols introduced by RD* $(\star)$	
$\mathcal{L} \subseteq \{0, 1, 2, 3\}$ $(\star)$	Target stage indices for reweighting
$\rho_{\text{var}} \in (0, 1)$ $(\star)$	Variance threshold for per-head subspace estimation (Eq. (**??**))
$k_{\ell,b,h} \in \{1, \ldots, d_h\}$ $(\star)$	Number of task-relevant PCA components for head h at block (ℓ, b), determined by ρ_{var} (Eq. (**??**))
$\boldsymbol{\lambda}_{\ell,b,h} \in \mathbb{R}^{d_h}$ $(\star)$	Learnable per-component weight vector for head h at block (ℓ, b); initialised via Eq. (28), optimised via Eq. (34
$\mathbf{C}_{\ell,b,h} \in \mathbb{R}^{d_h \times d_h}$	Empirical covariance matrix of $\widetilde{\mathbf{R}}_{\ell,b,h}$
$\boldsymbol{\Phi}_{\ell,b,h} \in \mathbb{R}^{d_h \times d_h}$	Orthogonal matrix of eigenvectors of $\mathbf{C}_{\ell,b,h}$
λ_j^{ev}	j-th eigenvalue of $\mathbf{C}_{\ell,b,h}$, $\lambda_1^{\text{ev}} \geq \cdots \geq \lambda_{d_h}^{\text{ev}} \geq 0$

Table 4. **Selezione delle teste più specializzate (Z-score coerenza).** Per brevità è riportata solo la prima descrizione selezionata dall'OMP; i risultati completi e i mapping tra indici assoluti e gerarchici sono consultabili nel notebook di analisi su GitHub.

Head ID	Layer	Dominant Cat.	Coherence Z	PC1 Var.	Primary Desc.
L3_B0_H7	3	*human_body*	1.57	0.13	baby crying
L2_B4_H1	2	*human_body*	1.46	0.11	person sneezing
L2_B5_H15	2	*nature*	1.23	0.14	water dripping
L3_B0_H25	3	*nature*	1.19	0.19	water dripping
L3_B1_H9	3	*urban*	0.79	0.16	chainsaw cutting
L2_B5_H6	2	*animals*	0.71	0.14	frog croaking
L3_B1_H17	3	*animals*	0.65	0.29	rooster crowing

Table 5. RD*hyperparameters and default values.

Symbol	Description	Default
$\mathcal{L}$	Target stage indices	$\{1, 2, 3\}$
ρ_{var}	Variance threshold for $k_{\ell,b,h}$ (Eq. (**??**))	0.95
η	Learning rate (Schedule-Free Adam)	10^{-2}
p	Early-stopping patience (epochs)	5
$T_{\max}$	Maximum optimisation epochs	30
$N_{\max}$	Maximum fitting clips	$10\,000$